Initially the rotor is rotated in low speed and after some time, the speed is increased step by step. Then check there is any vibration or sound in the motor.

10.6 Precautionary measures to be taken before using electrical machines

10.6.1 Alignment

When the motor shaft and load shaft are connected through the flange, note that the shafts are placed in one line. Plum bob, Try square and Dial guage are used to measure the alignment.

10.6.2 Airgap

The gap between stator and the rotor of a machine is called 'Air gap'. The air gap is increased when the bearing is worn out and depreciation is bearing housing. To measure air gap, feeler gauge is used.

10.6.3 Armature balancing

The rotor or armature of the machine is in roller shape. The weight of armature should be equal in all sides at all parts is called balancing. If weight is not equal then it is called unbalancing. i.e, one side of armature has more weight and the other side has less weight. This is called unbalanced rotor.

10.6.4 Importance of balancing

If the armature or rotor is not in balanced condition, the rotor will hit the stator and bearing gets damaged while rotating. If there is any damage in bearing, the alignment will get change. If the alignment is not proper, the air gap between stator and rotor will become unequal. Hence balancing is very important.

10.6.5 Types of balancing

1. Static balancing
2. Dynamic balancing

1. Static balancing

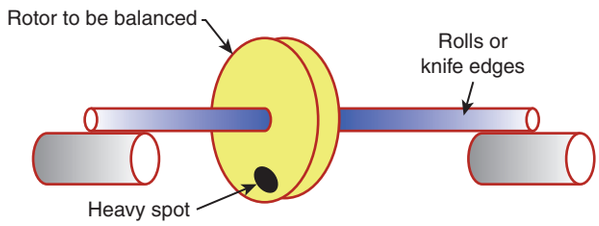


Fig. 10.1 Static balancing

The armature or rotor is placed on the two equal knife edges which are fitted on the flat surface. If the rotor is balanced it should not rotate. If it is unbalanced, it will stand in a slanting position. After one rotation is completed, The part or side which has more weight is placed in bottom side and the side which has light weight is placed on the upper side. Hence some weight should be added to the upper side for balance or reduce some weight from the lower side. Again do the test, if it is unbalance then add or remove the weight till it is balancing.

2. Dynamic balancing

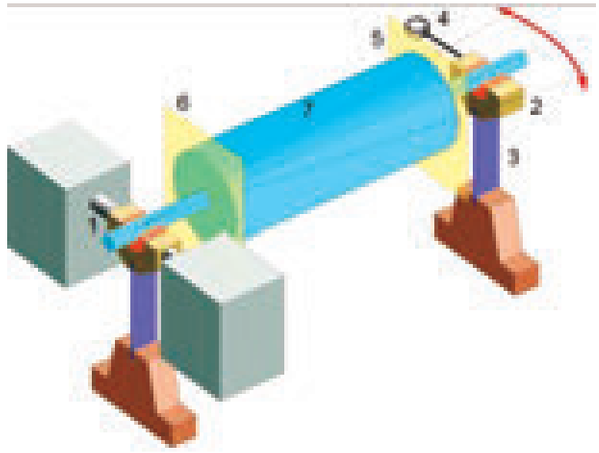


Fig. 10.2 Dynamic balancing

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